

HOMEWORK 9
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

CHECK: 04/26/2019

SECTION 5.4 - EIGENVECTORS AND LINEAR TRANSFORMATIONS

1. Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$ be the transformation that maps a polynomial, $p(t)$, to its derivative, $p'(t)$. Show that T is a linear transformation. Then, find the matrix for T relative to the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \{1, t\}$ for $\mathbb{P}_1$.
 2. The *trace* of an $n \times n$ matrix A is the sum of the diagonal entries in A and is denoted $\text{tr}(A)$. If B is another $n \times n$ matrix, can be shown that $\text{tr}(AB) = \text{tr}(BA)$. Use this fact to show that if A and B are similar, then $\text{tr}(A) = \text{tr}(B)$.
 3. Show that the trace of a diagonalizable matrix A is equal to the sum of the eigenvalues of A . (While we won't prove it here, this is actually true for *any* matrix, regardless of whether or not it is diagonalizable.)
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SECTION 5.5 - COMPLEX EIGENVALUES

5. Let the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$ act on $\mathbb{C}^2$. Find the eigenvalues of A and find a basis for each eigenspace of A in $\mathbb{C}^2$.
6. Let A be an $n \times n$ real matrix, and let $x \in \mathbb{C}^n$. Show that

$$\text{Re}(Ax) = A(\text{Re}(x)) \quad \text{and} \quad \text{Im}(Ax) = A(\text{Im}(x)).$$

SECTION 5.7 - APPLICATIONS TO DIFFERENTIAL EQUATIONS

5. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$. Determine whether the origin is a stable or unstable equilibrium point of the system $x'(t) = Ax(t)$.
6. Let A be as in problem 5. Solve the initial value problem

$$\begin{cases} x'(t) = Ax(t) \\ x(0) = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{cases}$$

Using your answer to problem 5, is it true that the solution you found satisfies

$$\lim_{t \rightarrow \infty} x(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} ?$$